

NEPAL FOOD COMPOSITION DATABASE



Government of Nepal
Nepal Agricultural Research Council
National Food Research Centre
Khumaltar, Lalitpur



Nepal Food Composition Database

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Best Wishes !!!

NARC Building
Singhdurbar Plaza
Kathmandu
NEPAL

I wish to extend my heartfelt congratulations for successfully publishing the book on Nepalese Food Composition Database. This epic work is not only a significant achievement of the National Food Research Centre (NFRC), Nepal Agricultural Research Council (NARC) but also a key resource for policymakers, nutritionists, researchers, academicians and farmers. By providing comprehensive data on the nutritional composition of Nepalese foods, this book will unquestionably contribute to better health outcomes, informed decision-making, and the promotion of local agricultural products.

The NFRC teamwork has been instrumental in completing this project, and I wish the team continued success in all future endeavours. May this publication pave the way for further research and innovation, ultimately enhancing food and nutrition security in Nepal. I would also like to thank the Asian Food and Agriculture Initiative Cooperation (AFACI), and Rural Development Administration (RDA), South Korea for their financial support in bringing this publication, and hope for further collaboration in the future.

Once again, congratulations to the entire team of NFRC for this remarkable achievement.

Dr. Doj Raj Khanal
Sept. 18, 2024

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**Khumaltar, Lalitpur
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FOREWORD

It is with great pride of accomplishment that we present the "Nepalese Food Composition Database." This publication represents a significant milestone in our ongoing efforts to enhance food security, nutritional awareness, and public health in Nepal. The accurate and detailed data contained within these pages are the result of extensive research, collaboration, and dedication from a multidisciplinary team of experts.

Understanding the nutritional composition of our food is essential for numerous stakeholders, including policymakers, healthcare professionals, researchers, and the general public. This database serves as a critical resource, providing scientifically validated information on the nutritional content of foods commonly consumed in Nepal. It is an indispensable tool for designing effective nutritional programs, conducting research, and formulating policies that address the unique dietary needs of our population.

I would like to thank Mr. Pravin Ojha (Scientist-S2) and his team for their continuous effort to make this happen. I would also like to thank the Asian Food and Agriculture Initiative Cooperation (AFACI), and Rural Development Administration (RDA), South Korea for their continuous support (technical and financial) throughout the project. It is my sincere hope that the "Nepalese Food Composition Database" will be widely utilized and appreciated by all those engaged in the field of food and nutrition and that it will inspire further advancements in our understanding of the foods that sustain us.

.....
Dr. Roman Karki
Chief

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Abbreviations

NARC	Nepal Agricultural Research Council
NFRC	National Food Research Centre
AFACI	Asian Food and Agriculture Initiative Cooperation
RDA	Rural Development Administration
I/NGOs	International /Non-Governmental Organizations
USDA	United States Department of Agriculture
CoFID	Composition of Food Integrated Dataset
FSANZ	Food Standards Australia New Zealand
IFCT	Indian Food Composition Tables
ICMR	Indian Council of Medical Research
FAO/INFOODS	Food and Agriculture Organization/International Network of Food Data System
MoALD	Ministry of Agriculture and Livestock Development
DFTQC	Department of Food Technology and Quality Control
DHS	Department of Health Service
AOAC	Association of Official Agricultural Chemists
ASTM	American Society for Testing and Materials
L/B Ratio	Length/Breadth Ratio
1000 KW	1000 Kernel Weight
EN	Energy
Prot	Protein
CF	Crude fiber
Carb	Carbohydrate
RS	Reducing Sugar
Ca	Calcium
Fe	Iron
P	Phosphorus
K	Potassium
Na	Sodium
Zn	Zinc
TC	Total Carotenoids
Vit C	Vitamin C

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Chapter 1

Food Composition Database

1.1 Introduction to Food Composition Database

The detailed database of the nutritional information of various foods including energy, macronutrients, and micronutrients is called the food composition database. The information must be quantitative. It must include a large set of nutritional information about food that is useful to the users. The users may be policymakers, researchers of food science, technology, and diet, public health workers, dieticians, food manufacturing industry, I/NGOs, consumers, academicians, and farmers, however, the application might be different for different users.

The different countries have their food composition database that reflects their own food and consumption patterns. These include

a. USDA FoodData Central

This food database is managed by the Agriculture Research Service, USDA. This database includes different categories of food such as foundation food (Commodity and minimally processed food), experimental food (sourced from researchers who work in collaboration with USDA and is published in peer-reviewed journals), branded foods (public-private partnership with USDA), etc. USDA also published a database for dietary studies.

b. McCance and Widdowson's Composition of Foods

This food database is published by Public Health England, UK government. They named it as Composition of Food Integrated Dataset (CoFID). The food is classified into different food groups.

c. Australian Food Composition Database

This Australian food composition database is managed by Food Standards Australia New Zealand (FSANZ). This includes food from cereals to seafood and includes all types of data which were analyzed, borrowed, and calculated.

d. Standard Tables of Food Composition Database

The Standard Tables of Food Composition Database of Japan is published by the Ministry of Education, Culture, Sports, Science, and Technology, Japan. The food composition database includes the food composition database of most of the raw and processed foods available in Japan.

e. Indian Food Composition Tables (IFCT)

The Indian Food Composition Tables is published by the National Institute of Nutrition, Indian Council of Medical Research (ICMR). The foods selected were 75% of the daily intake of the Indian Population.

f. China Food Composition

The Chinese Food Composition Tables is managed by the Institute of Nutrition and Food Safety, Chinese Academy of Preventive Medicine, which includes more than 600 foods.

g. Korean Food Composition Table

The Korean Food Composition Database is managed by the Rural Development Administration (RDA), this database provides detailed nutritional information on Korean foods, including traditional dishes like kimchi and bulgogi.

h. Thai Food Composition Database

The Thai Food composition database was developed by the Institute of Nutrition, Mahidol University, and reviewed using FAO/INFOODS Guidelines for checking food composition data before the Publication of a User Table.

i. Malaysian Food Composition Database

This database is managed by the Institute for Medical Research which includes the composition of diverse Malaysian food including indigenous foods and raw materials.

j. Food Composition Database Nepal

The food composition database of Nepal includes 16 groups of food based on commodities, cooked foods, etc. However, the food composition database lacks the sample collection area, sample scientific name, and other sample information.

Importance of Food Composition Database

- The food composition database is utilized for assessing the dietary intake and also planning the diet for individuals and groups.
- The government uses a food composition database to develop public health policies and dietary guidelines, address nutritional deficiencies, and agricultural planning.
- The food composition database is used for nutritional labels, making consumers informed about the nutritional profile.
- The study can be carried out to see the impact of nutritional intake and health outcomes.
- This database helps consumers to make their dietary choices.
- The database is used by the industry to develop new products with better nutritional profiles and develop nutritional supplements, fortified foods, and specialized foods required by sports persons.
- The food composition database also explores the dietary and culinary diversity around the nation and globe.
- The food composition database is used by agricultural researchers to study the effect of input on the quality of food and plan accordingly to improve food nutritional quality.

Limitations of Food Composition Database

- The information on nutrients may be less or may lack information on certain foods limiting the accuracy of dietary assessment and research.
- The variation due to variety, soil, inputs, and geographical variations may not be addressed by a food composition database.
- The food composition database if not updated, then it may lack variation due to changes in agricultural practices, not including the profile of new foods, and food fortification.
- The food composition database may include data on foods that are collected from the region near the laboratory or have easy access, so sometimes lacks information on diverse food.
- The various methods of analysis protocol, reporting nutrient content, and also the technical strength of laboratories vary around the globe, which results in a lack of standardization around the food composition database of different countries.

Factors affecting the nutritional value of food

- The nutrient profile of food is affected by geographical diversity such as soil type, climate, etc.
- Seasonal variation also affects the food composition.
- Agricultural practices such as irrigation practices, fertilizer employed, etc also affect the nutrient content.

- The variety and cultivar differences also affect the nutritional composition of food.
- The maturity stages of food also affect the nutrient quality of food.
- The sampling methodology, sample handling, and storage also affect the nutritional value of food.

1.2 Methodology for Development of Food Composition Database

Development of Steering Committee

The steering committee is formed including seven members, which is approved by the Executive Director of Nepal Agricultural Research Council. The steering committee members include:

Table 1.1 Steering Committee members

Members	Office
Planning Director (Dr. Luma Nidhi Pandey), Principal Scientist (S-5)	NARC
Chief, National Food Research Centre (Dr. Roman Karki), Senior Scientist (S-4)	NARC
Planning Chief (Dr. Tek Prasad Gotame), Senior Scientist (S-4)	NARC
Principle Investigator, National Food Research Centre (Mr. Pravin Ojha), Scientist (S-2)	NARC
Mr. Pramod Koirala, Senior Food Research Officer	Ministry of Agriculture and Livestock Development (MoALD)
Dr. Huma Bokkhim, Senior Food Research Officer	Department of Food Technology and Quality Control (DFTQC)
Mrs. Meena Mote, Senior Community Nursing Officer	Department of Health Service (DHS)

The steering committee members' meeting is held a minimum thrice in a year. The objective of the steering committee formation is:

- The first objective of the steering committee is to establish the goal for the database such as intended users, types of food, and nutrients to be included.
- The second objective of the steering committee is to set standards for data collection and analysis methods to ensure the reliability and consistency of data.
- The third objective is to monitor the process of data composition database development including sample collection to data management, reflecting the current food composition database.

The sample collection basis and guidelines are made, which are as follows:

- The sample should be raw food.
- The sample should be on stage ready to prepare, cook, or process, i.e. maturity stage ready for harvest.
- The sample can be a registered variety or a local variety.
- The sample should be collected from three plots, plants, or trees, and mixed to homogenize to counteract the variation and make it more representative.
- The sample site selection should be done based on the availability of resources such as humans and other sampling equipment along with the distance from the location of the laboratory.

- For cereals, legumes, potatoes, and tubers, samples can be collected in a cotton bag or net bag.
- For vegetables and fruits, if a sample can be collected and transferred to the laboratory within 2-4 hours then the sample can be collected in a net bag, otherwise, the sample should be collected and transferred in an icebox.

The samples collected and analyzed in three years were as follows:

Table 1.2: Sample Collected based on Commodity-wise.

Cereals	Legumes	Potato and tubers	Fruits	Vegetable
Rice-9	Soybean-2	Potato-3	Apple-2	Cauliflower-2
Wheat-6	Black gram	Roots-5	Citrus-4	Leafy vegetables-9
Maize-5	Chickpea	Sweet potato-3	Mango-2	Salad vegetables-4
Buckwheat-2	Beans-2		Litchi-2	Chilli
Millet-5	Lentil		Gooseberry	Brinjal
Barley-2	Mung		Banana	Gourd vegetable-3
Amaranth	Horse gram-2		Papaya	Pumpkin-2
Proso millet-2	Kidney beans-2		Guava	Tomato-2
Foxtail millet			Strawberry	Chayote
Sorghum			Watermelon	<i>Tane Bodi</i>
			Pear	
34	12	11	17	26

The figure of some sample collected is shown in following figure:

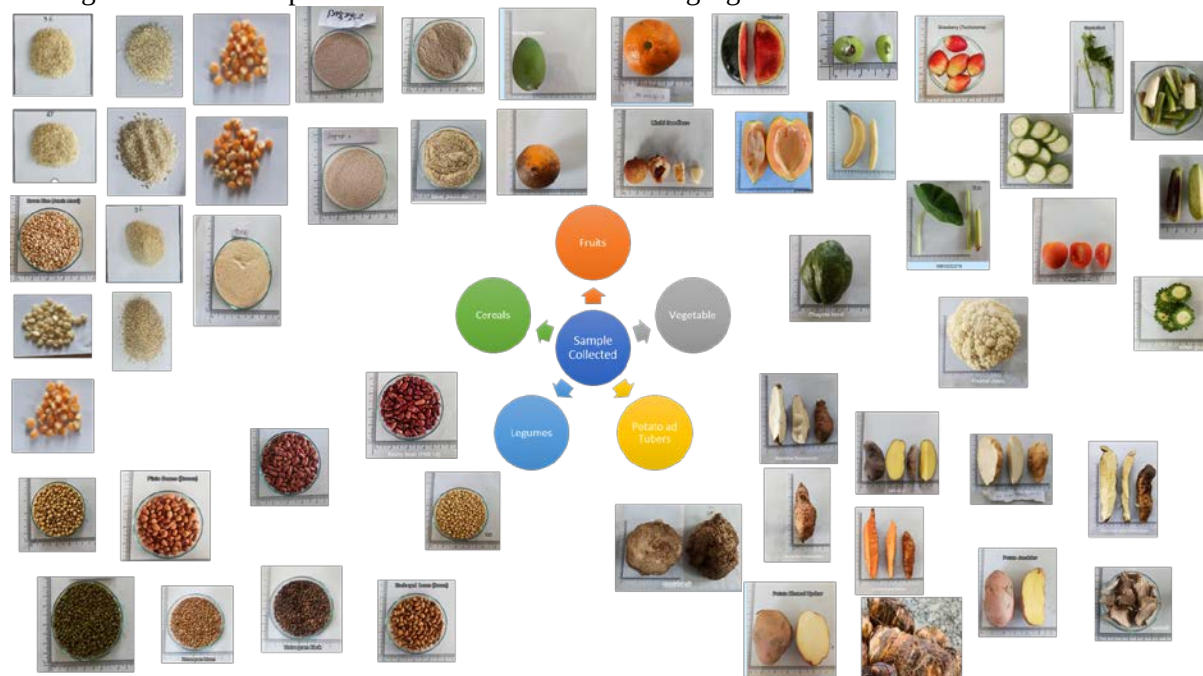


Figure 1.1 Sample collected

The analysis methodology chosen is as follows:

Table 1.3 Nutrient selected for analysis

No.	Nutrients	Analysis Method	Reference
1	Energy	The energy liberated while burning food by the Bomb Calorimeter	ASTM E711-87 (ASTM, 2004)
2	Water	Hot air oven method (105°C), Dry oven	AOAC 930.15 (AOAC, 2005)
3	Protein	Kjeldahl method, Micro distillation, Multiplying Nitrogen by 6.25	AOAC 920.152 (AOAC, 2005)
4	Fat	Soxhlet apparatus	AOAC 991.36 (AOAC, 2005)
5	Ash	Muffle (550°C)	AOAC 945.46 (AOAC, 2005)
6	Carbohydrate	By different methods, Subtracting water, protein, fat, ash, and crude fiber from 100	
7	Calcium	Titrimetric methods	AOAC 986.25 (AOAC, 2005)
8	Iron	Spectrophotometric methods	AOAC 944.02 (AOAC, 2005)
9	Phosphorus	Spectrophotometric methods	AOAC 991.25 (AOAC, 2005)
10	Potassium	AAS	AOAC 990.23 (AOAC, 2005)
11	Sodium	AAS	AOAC 990.23 (AOAC, 2005)
12	Zinc	AAS	AOAC 984.27 (AOAC, 2005)
13	Vitamin C	Titrimetric methods	AOAC 967.21 (AOAC, 2005)

The sampling location according to the province is listed in the following tables:

Table 1.4: Sample collected province wise

Province	Cereals	Legumes	Potato and Tubers	Fruits	Vegetables
Koshi			1	3	2
Madhesh	6		1	5	1
Bagmati	15		6	3	22
Gandaki	3	4	2		1
Lumbini	5	8		1	
Karnali	4			2	
Sudurpaschim	1		1	3	

The sample collected is shown in the following figure:

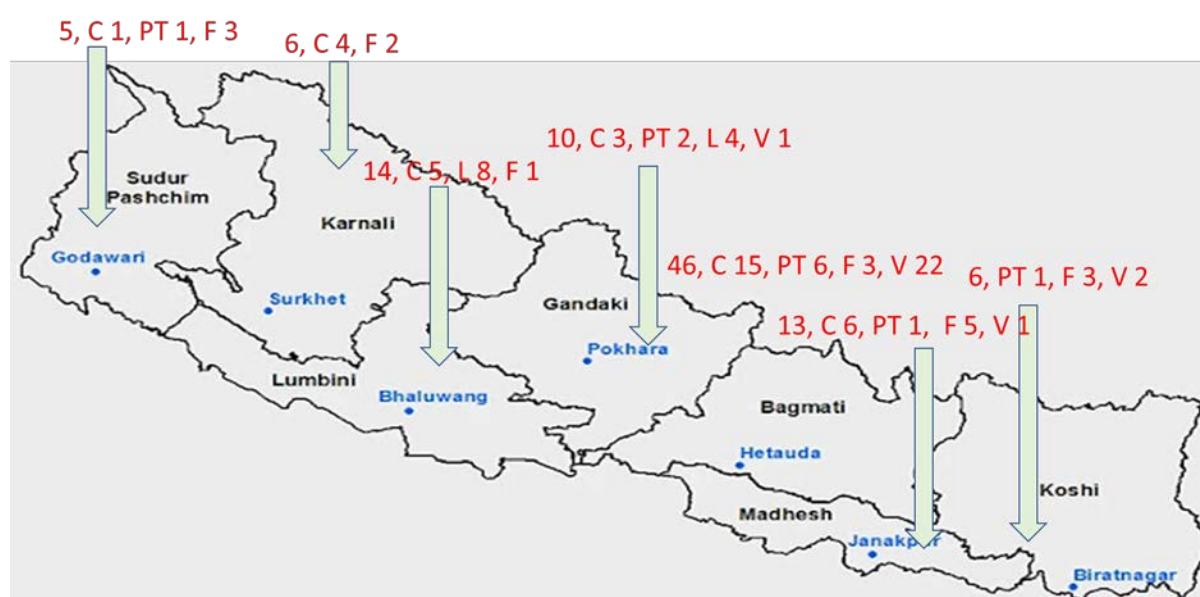


Figure 1.2 Collection of sample province-wise

The overall framework of the food composition database is as follows:

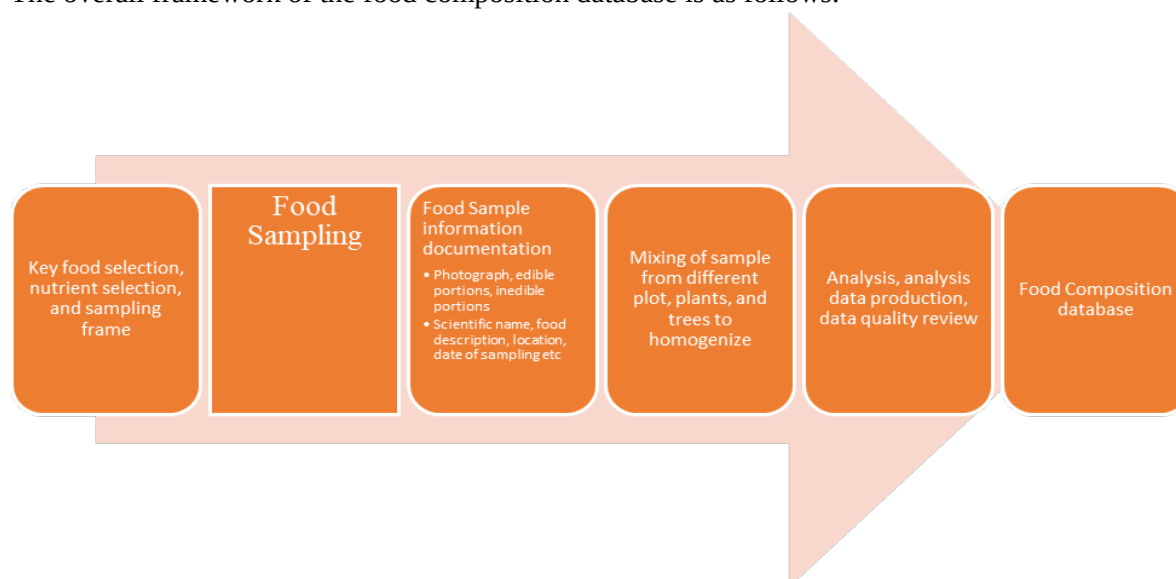


Figure 1.3: Overall framework of food composition database

Chapter 2

Nutritional Profile of Cereals and Pseudocereals

2.1 Cereals and pseudocereals: Product Information

Cereals and pseudocereals are important sources of carbohydrates, fibers, minerals, etc, however, the latter are broad-leaf plants while the former have grass-like characteristics. Cereals are true grasses while pseudocereals are not true grass, so the family is also different. Cereals belong to the Poaceae family while pseudocereals belong to other families like Polygonaceae, Amaranthaceae, etc. The most common examples of cereals are rice, wheat, maize, etc, while pseudocereals are buckwheat, amaranth, quinoa, etc.

2.1.1 Cereals: Rice

Rice belongs to the family Poaceae, genus *Oryza*, and species *sativa*, mostly found in Asia and Nepal. However, some species are *glaberrima* found in African countries. Rice grain consists of the husk (outer layer), bran (the layer beneath the hull), endosperm (major part), and embryo (nutrient-rich germ). Rice is the staple crop in Nepal. Rice is a good source of carbohydrates, and vitamin B like thiamine, and riboflavin, and also contributes to protein (limited in certain essential amino acids like lysine). Rice can be classified based on size (extra long-over 7 mm, long-6.0 to 6.9 mm, medium-5.0 to 5.99 mm and short-less than 5.0 mm), based on shape (L/B ratio) (slender-over 3, quasi-slender-2.4 to 3, bold-2 to 2.3, and round-less than 2), based on grain weight (giant-over 23 mg/grain, big-18.1 to 23 mg/grain, small-12 to 18 mg/grain, and tiny-less than 12 mg/grain), based on amylose content (waxy-1 to 2%, very low amylose-2 to 9%, low-10-20%, intermediate-20-25%, and high-25 to 30%), and based on gelatinization temperature (low-55-69° C, Intermediate-70-74° C, and high-75-79° C).

Table 2.1: Physical Properties of Rice.

No.	Variety name	Collection area	Physical		Edible/inedible part	
			L/B Ratio	1000 KW (g)	Edible part (g/100g)	Inedible part (g/100g)
1	Rice, <i>Pokhrel's Basmati</i>	<i>Pokhara</i>	3.31	19.18	64.96	35.03
2	Rice, <i>Jetho budho</i>	<i>Pokhara</i>	3.32	19.85	69.16	30.83
3	Rice, <i>Swarna Sub 1</i>	<i>Jankapur</i>	3.25	19.45	72.33	27.66
4	Rice, <i>Samba Mansuli sub 1</i>	<i>Jankapur</i>	3.61	14.31	70.71	29.28
5	Rice, <i>Bahuguni 1</i>	<i>Jankapur</i>	4.38	23.52	67.04	32.95
6	Rice, <i>Chaite 5</i>	<i>Jankapur</i>	3.96	17.64	70.18	29.81
7	Rice, <i>Hardinath 3</i>	<i>Jankapur</i>	4.32	21.93	65.84	34.15
8	Rice, <i>Hardinath 6</i>	<i>Jankapur</i>	3.79	14.51	71.39	28.60
9	Rice, <i>Jumli Marsi</i>	<i>Jumla</i>	2.32	27.3	82.05	17.94



Paddy



Polished Rice

Figure 2.1 Rice, *Pokhrel Basmati*



Paddy



Polished Rice

Figure 2.2 Rice, *Jetho budho*



Paddy



Polished Rice

Figure 2.3 Rice, *Swarna Sub 1*



Paddy



Polished Rice

Figure 2.4 Rice, *Samba mansuli sab-1*



Paddy



Polished Rice

Figure 2.5 Rice, *Bahuguni 1*



Paddy



Polished Rice

Figure 2.6 Rice, *Chaite 5*



Paddy



Polished Rice

Figure 2.7 Rice, *Hardinath 3*



Paddy



Polished Rice

Figure 2.8 Rice, *Hardinath 6*



Paddy



Polished Rice

Figure 2.9 Rice, *Jumli marsi*

2.1.2: Cereals: Maize

Maize/Corn are also the staple crops of most parts of the world mainly in African and Latin American countries. It is one of the industrially important crops for both food and feed. Maize (*Zea mays*) belongs to the family Poaceae. The maize kernel consists of a pericarp as the outermost layer, then the aleurone layer closing the endosperm, and germ, which is located at one end of the kernel, near the tip cap. Maize is a good source of carbohydrates, dietary fiber, minerals, and vitamins however it lacks essential amino acids like lysine and tryptophan. Based on color, maize is yellow, white, and red, but a few varieties with blue and purple colors are also available in the market. Based on variety, maize is classified as dent corn, flint corn, sweet corn, popcorn, flour corn, and pod corn.

Table 2.2: Physical Properties of Maize.

No.	Variety name	Collection area	Physical	Edible/inedible part	
			1000 Kernel weight (g)	Edible part (g/100g)	Inedible part (g/100g)
1	Maize, <i>Deuti</i>	Chitwan	357.50	100	0
2	Maize, <i>Manakamana-3</i>	Chitwan	366.49	100	0
3	maize, <i>Rampur composite</i>	Chitwan	304.7	100	0
4	Maize, <i>Arun 2</i>	Chitwan	268.9	100	0
5	Maize, <i>Mankamana 4</i>	Chitwan		100	0



Figure 2.10 Maize, *Deuti*



Figure 2.11 Maize, *Manakamana-3*



Figure 2.12 Maize, *Rampur composite*



Figure 2.13 Maize, *Arun-2*



Figure 2.14 Maize, *Manakamana*

2.1.3: Cereals: Wheat

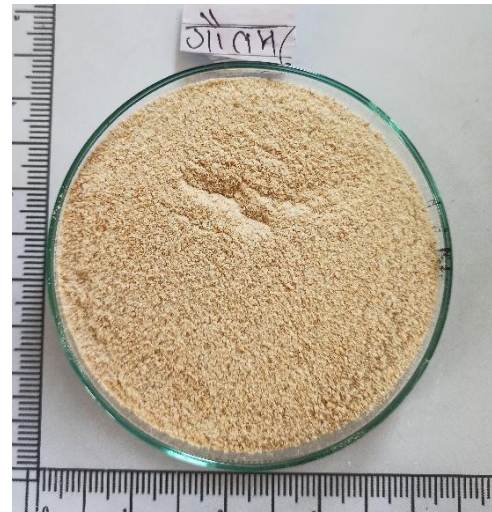
Wheat also serves as a staple food to a large population of the globe. The major product of wheat is flour, which is used for making baked products like bread, biscuits, cake, etc., extruded products like pasta and macaroni, and also used in houses in South Asia to make flat bread (*roti*). It belongs to the family Poaceae and the genus *Triticum*. The species are *aestivum* (bread wheat), *durum* (durum wheat), *turgidum* (Emmer wheat), and *spelta* (spelt wheat). Wheat is also classified as winter wheat, which grows in winter, and spring wheat, which grows in late summer. Similarly, wheat based on protein content is classified as hard wheat (high protein) and soft wheat (low protein). Wheat is a good source of carbohydrates, protein, dietary fiber, and minerals. However, the problems associated with wheat are celiac disease (gluten intolerance) and high glycemic index when refined flour is consumed.

Table 2.3: Physical Properties of Wheat.

No.	Variety name	Collection area	Physical		Edible/inedible part	
			L/B Ratio	1000 KW (g)	Edible part (g/100g)	Inedible part (g/100g)
1	Wheat, <i>Gautam</i>	<i>Bhairawa</i>	2.05	36.24	86.23	13.76
2	Wheat, <i>Bijaya</i>	<i>Bhairawa</i>	1.86	48.44	92.98	7.02
3	Wheat, <i>Mudule</i>	<i>Lalitpur</i>	2.14	44.2	92.33	7.66
4	Wheat, Zinc wheat 1	<i>Bhairawa</i>	1.98	42	95.15	4.84
5	Wheat, Zinc wheat 2	<i>Bhairawa</i>	1.91	41	94.65	5.34
6	Wheat, <i>Borlog</i>	<i>Bhairawa</i>	2.25	44	94.56	5.43



Wheat

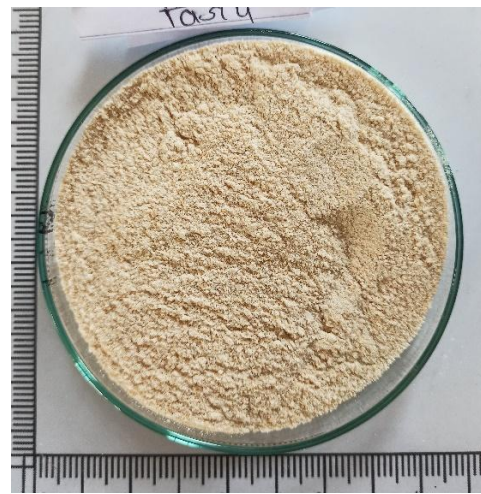


Flour

Figure 2.15 Wheat, *Gautam*



Wheat



Flour

Figure 2.16 Wheat, *Bijaya*



Wheat



Flour

Figure 2.17 Wheat, *Mudule*

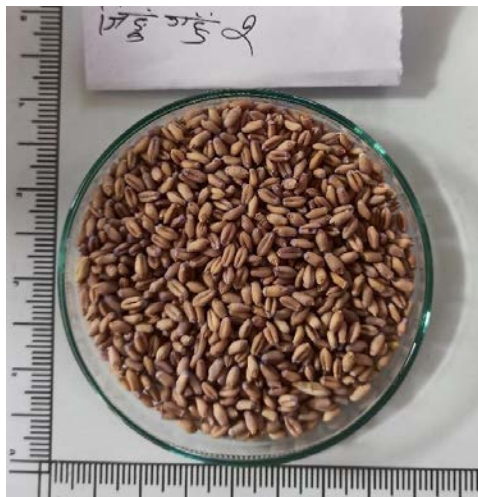


Wheat



Flour

Figure 2.18 Wheat, Zinc Wheat-1



Wheat



Flour

Figure 2.19 Wheat, Zinc Wheat-2



Wheat



Flour

Figure 2.20 Wheat, *Borlong*

2.1.4: Cereals: Millet

Millet is also considered a future smart crop due to low water requirement and is the fourth largest production crop after rice, maize, and wheat. Millet also belongs to the grass family (Poaceae) but in a different genus. The major millet found in Nepal are finger millet (*Eleusine coracana*), foxtail millet (*Setaria italica*), and proso millet (*Panicum miliaceum*). There are other millets also found in Nepal namely sorghum, barnyard/sanwa millet, pearl millet, little millet, and kodo millet. Millet is also used for food and feed, and due to its health implications in non-communicable diseases, people are now more searching for millet. However, processing difficulties make it difficult for industrial applications in countries like Nepal. Millet is also a good source of carbohydrates, protein, dietary fiber, minerals, and vitamins.

Table 2.4: Physical Properties of Millet.

No.	Variety name	Scientific name	Collection area	Physical		Edible/inedible part	
				L/B Ratio	1000 KW (g)	Edible part (g/100g)	Inedible part (g/100g)
1	Finger millet, <i>Dalle kodo 1</i>	<i>Eleusine coracana</i>	<i>Dolakah</i>	1.01	2.2	84.56	15.44
2	Finger millet, <i>Sailung kodo 1</i>	<i>Eleusine coracana</i>	<i>Dolakah</i>	1.01	3.1	83.28	16.72
3	Finger millet, <i>Okhle kodo 1</i>	<i>Eleusine coracana</i>	<i>Dolakah</i>	0.98	2.9	83.76	16.24
4	Finger millet, <i>Kabre kodo 2</i>	<i>Eleusine coracana</i>	<i>Dolakah</i>	1.01	1.63	82.66	17.34
5	Finger millet, <i>Kabre kodo 1</i>	<i>Eleusine coracana</i>	<i>Dolakah</i>	1.06	2.68	85.18	14.82
6	Proso millet, <i>Dudhe chino</i>	<i>Panicum miliaceum</i>	<i>Jumla</i>	1.28	7.4	79.86	20.14
7	Proso millet, <i>Hade chino</i>	<i>Panicum miliaceum</i>	<i>Jumla</i>	1.30	6.44	78.57	21.43
8	Foxtail millet, <i>Bariyo kaguno</i>	<i>Setaria italica</i>	<i>Lamjung</i>	0.97	1.77	78.57	21.43



Figure 2.21 Millet, *Dalle kodo 1*



Figure 2.22 Millet, *Shailungkodo 1*



Figure 2.23 Millet, *Okhley kodo 1*



Figure 2.24 Millet, *Kavre kodo 2*



Figure 2.25 Millet, *Kavre kodo 1*



Figure 2.26 Proso Millet, *Dudhe chino*



Figure 2.27 Proso Millet, *Hade chino*



Figure 2.28 Foxtail, Millet, *Bariyo kaguno*

2.1.5: Cereals: Barley, amaranth, buckwheat, and sorghum

Barley has highlighted its importance as one of the major industrial crops for both food and feed. Barley (*Hordeum vulgare* L.) is an important hill crop of Nepal. Although its use has been common in worship and death rituals, it is also used to prepare different types of food items. Barley is classified as either six-row (6R) or two-row (2R), depending on the physical arrangement of the kernels on the plant. Two-row barley has lower protein content than Six-row barley and is therefore more suitable for malt production. High-protein barley is best suited for animal feed. Barley is also described as hulled or hull-less depending on the presence of beards or awns covering the kernels. The barley is also classified into winter and spring. Barley belongs to the family Poaceae. They are a good source of carbohydrates, dietary fibers, and minerals.

Amaranth belongs to the family amaranthaceae and the genus amaranthus. They are a good source of carbohydrates, protein, dietary fiber, and minerals. The amino acid composition of amaranth is far better than the cereals from the Poaceae family. Some species are also used for leafy vegetables in Nepal.

Buckwheat belongs to the Polygonaceae family, and the most common are *Fagopyrum esculentum* (common buckwheat) and *Fagopyrum tataricum* (Tartary buckwheat). They are also a good source of carbohydrates, protein, dietary fibers, and minerals. Due to the flavonoid (rutin), it is considered to be a healthy food with medicinal value for cardiovascular benefits. They also contain all essential amino acids that differentiate them from grass family crops.

Sorghum belongs to the family Poaceae. The scientific name is *Sorghum bicolor*. They are staple crops in some African countries, and also have industrial applications in distillery, and are also used in feed. Sorghum is classified as grain sorghum, sweet sorghum, broomcorn sorghum, etc based on their application. The gluten-free grain is rich in carbohydrates, protein, dietary fibers, and minerals.

Table 2.5: Physical Properties of Cereals

No.	Variety name	Scientific name	Collection area	Physical		Edible/inedible part	
				L/B Ratio	1000 KW(g)	Edible part (g/100g)	Inedible part g/100g
1	Barley, Bonus	Hordeum vulgare	Dolakha	2.66	36.89	75.34	24.66
2	Naked barley, Solu uwa	Hordeum vulgare	Dolakha	1.98	31.92	82.45	17.55
3	Amaranth	Amaranthus cruentus	Doti	1.48	0.31	100.00	0
4	Buckwheat	Fagopyrum esculentum	Dolakha	1.71	12.8	100.00	0
5	Tartary Buckwheat	Fagopyrum tataricum	Dolakha	1.80	17.39	70.59	29.41
6	Sorghum	Sorghum bicolor L. Moench	Dolakha	1.11	19.66	91.84	8.16



Figure 2.29 Barley, Bonus



Figure 2.30 Naked Barley, Solu uwa



Figure 2.31 Amaranth



Buckwheat



Flour

Figure 2.32 Tartary Buckwheat



Buckwheat



Flour

Figure 2.33 Buckwheat



Sorghum



Flour

Figure 2.34 Sorghum

2.2 Cereals and pseudocereals: Nutritional Information

2.2.1: Cereals: Rice

Table 2.6: Energy, proximate, and reducing sugar of 100 g of rice (edible part).

No.	Food and Description	Energy kcal	Water g	Prot g	Fat g	Ash g	CF g	Carb g	RS g
1	Rice, <i>Pokhrel</i> basmati	318	14.29	8.43	1.00	0.57	1.36	73.03	0.17
2	Rice, <i>Jetho budho</i>	324	13.87	7.19	1.25	0.68	0.96	74.05	0.15
3	Rice, <i>Swarna sab-1</i>	325	13.05	7.19	0.89	0.59	1.03	75.35	0.23
4	Rice, <i>Samba mansuli sab-1</i>	325	13.32	7.34	1.08	0.60	1.14	74.52	0.31
5	Rice, <i>Bahuguni 1</i>	327	13.11	8.32	0.71	0.46	0.97	74.44	0.42
6	Rice, <i>Chaite 5</i>	320	13.05	7.58	0.92	0.53	1.44	74.48	0.48
7	Rice, <i>Hardinath 3</i>	326	12.96	8.42	0.88	0.54	0.89	73.97	0.34
8	Rice, <i>Hardinath 6</i>	328	13.01	8.29	0.90	0.54	0.95	74.31	0.22
9	Rice, <i>Jumli marsi</i>	328	11.87	9.81	2.41	1.31	2.52	72.08	0.27

*Note: CF=Crude fiber, Prot= Protein, Carb= Carbohydrate, RS= Reducing Sugar

Table 2.7: Minerals and vitamin content of 100 g of rice (edible parts).

No.	Food and Description	Ca mg	Iron mg	P mg	K mg	Na mg	Zinc mg	TC µg	Vit C mg
1	Rice, <i>Pokhrel</i> basmati	12.23	2.43	118.67	137.29	2.47	1.28	30.67	ND
2	Rice, <i>Jetho budho</i>	12.17	1.99	91.00	134.60	2.50	1.23	34.67	ND
3	Rice, <i>Swarna sab-1</i>	19.15	2.30	65.67	118.14	2.33	1.09	19.33	ND
4	Rice, <i>Samba mansuli sab-1</i>	17.58	2.30	65.67	111.31	2.18	1.05	16.33	ND
5	Rice, <i>Bahuguni 1</i>	12.31	2.51	119.00	122.38	2.31	1.16	20.00	ND
6	Rice, <i>Chaite 5</i>	11.43	2.13	141.67	120.05	2.09	1.13	21.33	ND
7	Rice, <i>Hardinath 3</i>	14.39	3.11	57.67	116.91	2.09	1.12	24.00	ND
8	Rice, <i>Hardinath 6</i>	11.58	2.44	177.67	113.41	2.17	1.08	15.00	ND
9	Rice, <i>Jumli marsi</i>	55.47	2.31	100.81	161.35	3.61	1.18	169.67	ND

*Note: Ca= Calcium, P= Phosphorus, K= Potassium, Na= Sodium, TC= Total Carotenoids, Vit C= Vitamin C

2.2.2: Cereals: Maize

Table 2.8: Energy, proximate, and reducing sugar of 100 g of maize (edible part).

No.	Food and Description	Energy kcal	Water g	Prot g	Fat g	Ash g	CF g	Carb g	RS g
1	Maize, <i>Deuti</i>	320.00	12.28	8.36	3.44	1.24	2.55	72.13	2.24
2	Maize, <i>Manakamana-3</i>	320.00	11.50	5.85	3.35	2.05	2.61	74.63	1.86
3	Maize, <i>Rampur composite</i>	330.00	12.50	7.69	4.02	0.12	2.41	73.26	1.94
4	Maize, <i>Arun 2</i>	334.03	9.61	8.50	3.92	1.50	2.57	73.90	2.12
5	Maize, <i>Mankamana</i>	329.96	11.45	8.57	4.47	1.17	2.62	71.73	1.64

*Note: CF=Crude fiber, Prot= Protein, Carb= Carbohydrate, RS= Reducing Sugar

Table 2.9: Minerals and vitamin content of 100 g of maize (edible parts).

No.	Food and Description	Ca mg	Iron mg	P mg	K mg	Na mg	Zinc mg	TC µg	Vit C mg
1	Maize, <i>Deuti</i>	14.22	1.74	252.27	283.10	4.26	2.27	842.00	ND
2	Maize, <i>Manakamana-3</i>	13.48	2.29	265.73	283.79	4.18	2.40	850.67	ND
3	Maize, <i>Rampur composite</i>	14.37	1.95	230.85	285.63	4.32	2.28	1134.00	ND
4	Maize, <i>Arun 2</i>	14.34	2.74	247.71	287.78	4.13	2.31	1162.00	ND
5	Maize, <i>Mankamana</i>	14.80	2.93	281.23	289.67	4.37	2.16	1247.67	ND

*Note: Ca= Calcium, P= Phosphorus, K= Potassium, Na= Sodium, TC= Total Carotenoids, Vit C= Vitamin C

2.2.3: Cereals: Wheat

Table 2.10: Energy, proximate, and reducing sugar of 100 g of wheat (edible part).

No.	Food and Description	Energy Kcal	Water g	Prot g	Fat g	Ash g	CF g	Carb g	RS g
1	Wheat, <i>Gautam</i>	316.00	10.71	11.64	1.62	2.50	2.33	71.19	1.11
2	Wheat, <i>Bijaya</i>	322.00	10.65	11.66	1.60	2.19	2.22	71.67	1.25
3	Wheat, <i>Mudule</i>	311.00	12.28	12.97	1.89	2.17	2.58	68.12	1.12
4	Wheat, Zinc wheat 1	312.00	12.63	11.88	1.95	2.47	2.26	68.81	1.23
5	Wheat, Zinc Wheat 2	308.00	13.35	12.24	2.26	2.14	2.60	67.41	1.19
6	Wheat, <i>Borlog</i>	311.00	13.05	11.79	2.26	2.45	2.42	68.03	1.33

*Note: CF=Crude fiber, Prot= Protein, Carb= Carbohydrate, RS= Reducing Sugar

Table 2.11: Minerals and vitamin content of 100 g of wheat (edible parts).

No.	Food and Description	Ca mg	Iron mg	P mg	K mg	Na mg	Zinc mg	TC µg	Vit C mg
1	Wheat, <i>Gautam</i>	69.86	3.00	143.31	314.02	2.07	1.93	286.67	ND
2	Wheat, <i>Bijaya</i>	67.64	3.22	154.55	312.78	1.98	1.92	323.33	ND
3	Wheat, <i>Mudule</i>	70.24	4.88	164.68	314.65	2.13	2.07	293.33	ND
4	Wheat, Zinc wheat 1	122.36	2.26	137.21	322.94	1.97	4.77	286.67	ND
5	Wheat, Zinc Wheat 2	114.05	4.90	143.24	294.86	2.13	4.75	326.67	ND
6	Wheat, <i>Borlog</i>	84.86	4.52	144.17	300.88	2.15	1.93	293.33	ND

*Note: Ca= Calcium, P= Phosphorus, K= Potassium, Na= Sodium, TC= Total Carotenoids, Vit C= Vitamin C

2.2.4: Cereals: Millet

Table 2.12: Energy, proximate, and reducing sugar of 100 g of millet (edible part).

Index	Food and Description	Energy kcal	Water g	Prot g	Fat g	Ash g	CF g	Carb g	RS g
1	Millet, <i>Dalle kodo 1</i>	278.00	13.18	7.67	1.11	2.22	4.09	71.74	0.78
2	Millet, <i>Sailung kodo 1</i>	305.00	11.87	7.35	0.96	2.20	2.61	75.02	0.69
3	Millet, <i>Okhle kodo 1</i>	288.00	12.78	7.82	1.10	2.35	3.56	72.40	0.71
4	Millet, <i>Kabre kodo 2</i>	289.00	12.97	7.51	1.06	2.06	3.53	72.86	0.76
5	Millet, <i>Kabre kodo 1</i>	283.00	13.00	7.15	0.95	2.34	3.85	72.72	0.68
6	Proso millet, <i>Dudhe chino</i>	309.00	10.96	10.69	4.51	3.54	3.49	66.81	0.46
7	Proso millet, <i>Hade chino</i>	305.00	11.97	12.63	4.50	3.43	4.82	62.66	0.40
8	Foxtail millet, <i>Bariyo kaguno</i>	288.00	11.27	11.02	4.02	2.71	6.31	64.67	0.48

*Note: CF=Crude fiber, Prot= Protein, Carb= Carbohydrate, RS= Reducing Sugar

Table 2.13: Minerals and vitamin content of 100 g of millet (edible parts).

Index	Food and Description	Ca mg	Iron mg	P mg	K mg	Na mg	Zinc mg	TC µg	Vit C mg
1	Millet, <i>Dalle kodo 1</i>	264.85	3.81	242.10	428.49	4.10	2.10	96.33	ND
2	Millet, <i>Sailung kodo 1</i>	233.70	4.35	238.14	431.77	4.16	2.31	84.67	ND
3	Millet, <i>Okhle kodo 1</i>	185.15	3.19	242.76	434.02	4.40	2.20	96.00	ND
4	Millet, <i>Kabre kodo 2</i>	218.73	3.84	235.68	445.00	3.97	2.94	108.00	ND
5	Millet, <i>Kabre kodo 1</i>	180.63	2.83	237.36	422.23	4.20	2.52	89.00	ND
6	Proso millet, <i>Dudhe chino</i>	284.40	3.68	284.79	413.55	3.84	2.17	135.33	ND
7	Proso millet, <i>Hade chino</i>	274.22	5.63	382.63	446.33	4.18	2.30	160.67	ND
8	Foxtail millet, <i>Bariyo kaguno</i>	41.88	2.55	284.67	251.99	4.63	2.63	100.33	ND

*Note: Ca= Calcium, P= Phosphorus, K= Potassium, Na= Sodium, TC= Total Carotenoids, Vit C= Vitamin C

2.2.5: Cereals: Barley, amaranth, buckwheat, and sorghum

Table 2.14: Energy, proximate, and reducing sugar of 100 g of Barley, amaranth, buckwheat, and sorghum (edible part).

Index	Food and Description	Energy	Water	Prot	Fat	Ash	CF	Carb	RS
		kcal	g	g	g	g	g	g	g
1	Barley, Bonus	281.00	11.66	8.40	1.69	1.32	6.86	70.08	2.63
2	Naked barley, <i>Solu uwa</i>	328.00	11.18	10.57	1.98	1.84	2.35	72.08	2.47
3	Amaranth, <i>Rato marsi</i>	331.00	10.64	14.48	5.09	2.29	3.61	63.89	0.76
4	Buckwheat, <i>Mithe fapar 1</i>	280.00	11.76	9.84	2.12	2.63	6.15	67.49	0.46
5	Tartary buckwheat	311.00	11.65	9.44	2.42	1.39	3.29	71.81	1.02
6	Sorghum	321.00	14.16	7.96	2.44	1.17	0.95	73.31	1.05

*Note: CF=Crude fiber, Prot= Protein, Carb= Carbohydrate, RS= Reducing Sugar

Table 2.15: Minerals and vitamin content of 100 g of Barley, amaranth, buckwheat, and sorghum (edible parts).

No.	Food and Description	Ca	Iron	P	K	Na	Zinc	TC	Vit C
		mg	mg	mg	mg	mg	mg	µg	mg
1	Barley, Bonus	33.74	1.91	205.73	253.05	6.39	1.66	102.33	ND
2	Naked barley, <i>Solu uwa</i>	25.69	2.62	187.90	180.39	4.77	1.99	94.67	ND
3	Amaranth, <i>Rato marsi</i>	176.91	7.46	453.93	385.49	2.76	2.66	121.67	ND
4	Buckwheat, <i>Mithe fapar 1</i>	52.89	6.43	385.25	431.62	2.50	3.28	223.33	ND
5	Tartary buckwheat	53.65	5.25	363.33	323.23	2.69	2.53	4421.33	ND
6	Sorghum	41.57	3.72	223.95	357.09	7.63	2.31	231.33	ND

*Note: Ca= Calcium, P= Phosphorus, K= Potassium, Na= Sodium, TC= Total Carotenoids, Vit C= Vitamin C

Chapter 3

Nutritional Profile of beans and legumes

2.1 Legumes

Legumes refers to the *Fabaceae* or *Leguminosae* family. Legumes are a good source of protein, dietary fiber, and minerals. Some legumes are eaten as seeds, while some are eaten as whole pods. Some beans are soaked before cooking while some are not soaked. The beans are consumed as vegetables, soups, salad, etc. depending upon the culinary practices of the community. In Nepal, legumes are consumed as *daal* (Split legumes, used to make thickened soup for consuming along with rice and *roti*), whole legumes (used to make thickened soup for consuming along with rice, beaten rice, and *roti*), roasted beans (as a snack), traditional food (*maseura*, *kinema*, etc), salad, and pickle.

Table 3.1: Physical analysis of Legumes.

S.No.	Food and Description		Collection area	Physical		Edible/inedible part	
	Variety Name	Scientific name		L/B Ratio	1000 KW (g)	Edible part (g/100g)	Inedible part (g/100g)
1	Soybean, <i>Puja</i>	<i>Glycine max</i>	<i>Banke</i>	1.37	15.74	100	0
2	Soybean, <i>Kawa</i>	<i>Glycine max</i>	<i>Banke</i>	1.14	105.13	100	0
3	Mung, <i>Pratikshya</i>	<i>Vigna radiata</i>	<i>Banke</i>	1.35	4.24	100	0
4	Chickpea, <i>Koseli</i>	<i>Cicer arietinum</i>	<i>Banke</i>	1.33	185.57	100	0
5	Red lentil, <i>Khajura masuro 3</i>	<i>Lens culinaris</i>	<i>Banke</i>	1.72	44.4	100	0
6	Red kidney bean, Local	<i>Phaseolus vulgaris</i>	<i>Doti</i>	2.05	308.1	100	0
7	Red Kidney bean, PDR 14	<i>Phaseolus vulgaris</i>	<i>Banke</i>	1.83	402.1	100	0
8	Black gram, Local	<i>Vigna mungo</i>	<i>Banke</i>	1.21	41.7	100	0
9	Horse gram, Local brown	<i>Macrotyloma uniflorum</i>	<i>Baglung</i>	1.48	29.5	100	0
10	Horse gram, Local black	<i>Macrotyloma uniflorum</i>	<i>Baglung</i>	1.48	37.3	100	0
11	String beans	<i>Phaseolus vulgaris</i>	<i>Gulmi</i>	1.38	17.49	100	0
12	Cow pea	<i>Vigna unguiculata</i>	<i>Gulmi</i>	1.45	37.38	100	0



Figure 3.1 Soybean, *Puja*



Figure 3.2 Soyabean, *Kawa*



Figure 3.3 Mung, *Pratikshya*



Figure 3.4 Chickpea, *Koseli*



Figure 3.5 Red lentil, *Khajura masuro 3*



Figure 3.6 Red kidney bean, *Local*

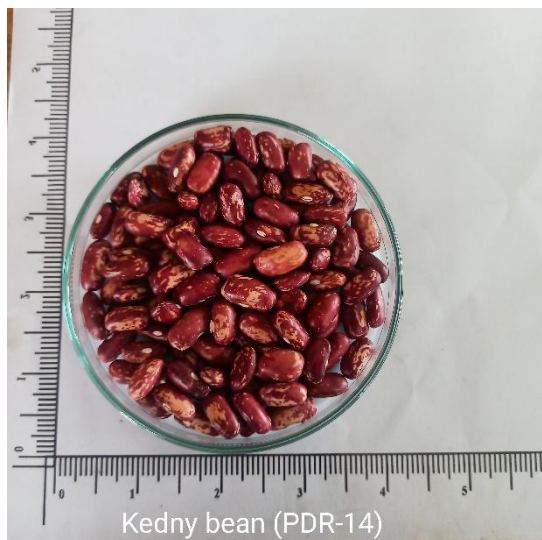


Figure 3.7 Red Kidney bean, PDR 14



Figure 3.8 Black gram, Local



Figure 3.9 Horse gram, Local brown



Figure 3.10 Horse gram, Local Black



Figure 3.11 String beans



Figure 3.12 Cow Pea

3.2 Legumes: Nutritional Information

3.2.1: Legumes

Table 3.2: Energy, proximate, and reducing sugar of 100 g of Beans and Legumes(edible parts)

Index	Food and Description	Energy kcal	Water g	Prot g	Fat g	Ash g	CF g	Carb g	RS g
1	Soybean, <i>Puja</i>	396.00	9.78	43.36	17.71	4.06	3.38	21.71	0.60
2	Soybean, <i>Kawa</i>	368.00	9.17	37.80	17.85	4.57	9.05	21.55	1.56
3	Mung, <i>Pratikshya</i>	329.00	9.52	27.16	0.45	2.41	1.62	58.84	0.55
4	Chickpea, <i>Koseli</i>	315.00	10.41	18.76	3.94	2.74	4.17	59.97	0.59
5	Red lentil, <i>Khajura masuro 3</i>	292.00	10.08	24.41	0.66	2.63	5.08	57.14	1.02
6	Red kidney bean, Local	258.00	9.40	19.98	1.73	3.45	8.28	57.16	1.43
7	Red Kidney bean, PDR 14	256.00	9.56	19.89	1.95	3.47	8.34	56.80	1.51
8	Black gram, Local	295.00	9.14	24.20	1.52	3.15	5.26	56.72	0.37
9	Horse gram, Local brown	295.00	10.58	22.56	0.42	3.56	3.20	59.69	0.20
10	Horse gram, Local black	294.00	12.04	22.60	0.58	2.70	3.18	58.90	0.17
11	String beans	226.00	11.46	16.99	0.72	3.17	8.05	59.61	0.15
12	cow pea	312.00	10.17	22.11	1.06	3.54	3.52	59.61	0.21

*Note: CF=Crude fiber, Prot= Protein, Carb= Carbohydrate, RS= Reducing Sugar

Table 3.3: Minerals and vitamin content of 100 g of Beans and Legumes(edible parts).

Index	Food and Description	Ca mg	Iron mg	P mg	K mg	Na mg	Zinc mg	TC µg	Vit C mg
1	Soybean, <i>Puja</i>	266.68	8.35	359.45	1453.31	2.12	3.85	393.67	ND
2	Soybean, <i>Kawa</i>	194.99	7.24	442.90	1519.91	2.42	3.35	228.67	ND
3	Mung, <i>Pratikshya</i>	220.48	3.25	331.48	1181.57	9.76	2.23	550.67	ND
4	Chickpea, <i>Koseli</i>	51.49	5.76	286.54	932.74	20.54	3.35	867.67	ND
5	Red lentil, <i>Khajura masuro 3</i>	57.45	6.31	294.52	765.78	9.99	3.46	767.33	ND
6	Red kidney bean, Local	136.05	5.84	386.59	1318.75	11.07	2.64	94.00	ND
7	Red Kidney bean, PDR 14	137.07	6.04	383.11	1323.96	15.29	2.73	78.67	ND
8	Black gram, Local	58.97	4.56	362.40	1149.34	17.06	2.94	216.33	ND
9	Horse gram, Local brown	220.48	3.25	331.48	1181.57	9.76	2.23	237.33	ND
10	Horse gram, Local black	283.58	6.33	331.48	1078.56	12.40	2.59	224.67	ND
11	String beans	108.52	4.76	277.31	1235.84	13.43	3.46	211.33	ND
12	cow pea	260.95	5.60	280.78	1123.02	12.02	2.46	221.33	ND

*Note: Ca= Calcium, P= Phosphorus, K= Potassium, Na= Sodium, TC= Total Carotenoids, Vit C= Vitamin C

Chapter 4

Nutritional Profile of Potato and Tubers

4.1 Potato and Tubers: Product Information

Potatoes are a specific type of tuber belonging to the family Solanaceae, whereas tubers are broad groups including potatoes, sweet potatoes, yam, cassava, etc. Tubers are rich in carbohydrates, minerals, vitamins, and dietary fibers. Potatoes are also a good source of Vitamin C. In the Himalayan region of Nepal, potatoes are not only a source of carbohydrates but protein also. There are many colored tubers, rich in bioactive components like carotenoids, anthocyanins, etc.

Table 4.1: Physical properties of Potato and Tubers.

No.	Food and Description	Scientific name	Collection area	Physical L/B Ratio	Edible/inedible part	
					Edible (g/100g)	Inedible part (g/100g)
1	Potato, MS-42-3	Solanum tuberosum	Lalitpur	2.33	81.71	18.29
2	Potato, <i>Janakdev</i>	Solanum tuberosum	Lalitpur	1.94	86.88	13.12
3	Potato, <i>Khumal upahar</i>	Solanum tuberosum	Lalitpur	1.17	97.72	2.28
4	Sweet potato, <i>Suntale 1</i>	Ipomoea batatas	Lalitpur	3.36	97.50	2.50
5	Sweet potato, Bensishar white	Ipomoea batatas	Lalitpur	3.39	97.31	2.69
6	Sweet potato, <i>Kimichaur seto</i>	Ipomoea batatas	Lalitpur	4.55	97.43	2.57
7	Elephant yam, Elephant yam	Amorphophallus paeoniifolius	Pokhara		89.69	10.31
8	Chayote root	Sechium edule	Dhankutta	2.04	93.63	6.37
9	Taro	Colocasia esculenta	Pokhara		80.64	19.36
10	Cassava	Manihot esculenta	Udaypur	8.41	84.74	15.26
11	Beet root, Local	Beta vulgaris L.	Kailali	0.98	87.33	12.67



Figure 4.1 Potato, MS-42-3



Figure 4.2 Potato, Janakdev



Figure 4.3 Potato, Khumal upahar



Figure 4.4 Sweet potato, Suntale 1



Figure 4.5 Sweet potato, Bensishar white



Figure 4.6 Sweet potato, Kimichaur seto



Figure 4.7 Elephant yam, Elephant yam



Figure 4.8 Chayote root



Figure 4.9 Taro



Figure 4.10 Cassava



Figure 4.11 Beet root, Local

4.2 Potato and Tubers: Nutritional Information

4.2.1: Potato and tubers

Table 4.2: Energy, proximate, and reducing sugar of 100 g of Potato and Tubers.

Index	Food and Description	Energy kcal	Water g	Prot g	Fat g	Ash g	CF g	Carb g	RS G
1	Potato, MS-42-3	81.00	77.98	1.28	0.35	0.70	0.78	18.91	0.46
2	Potato, <i>Janakdev</i>	64.00	82.33	1.33	0.08	0.73	0.59	14.95	0.41
3	Potato, <i>Khumal upahar</i>	71.00	80.57	1.59	0.08	0.73	0.55	16.48	0.36
4	Sweet potato, <i>Suntale 1</i>	102.00	70.45	1.21	0.38	1.08	1.28	25.60	0.89
5	Sweet potato, Bensishar white	112.00	69.25	0.89	0.29	0.95	0.88	27.74	0.87
6	Sweet potato, <i>Kimichaur seto</i>	106.00	70.11	1.33	0.23	1.30	0.97	26.05	0.85
7	Elephant yam, Elephant yam	79.00	76.73	0.78	0.10	1.15	1.02	20.22	0.47
8	Chayote root	106.00	72.21	0.62	0.14	1.26	0.42	25.35	0.43
9	Taro	83.00	73.86	1.47	0.28	1.49	1.04	21.87	1.36
10	Cassava	133.00	64.27	1.56	0.35	0.55	0.66	32.61	0.32
11	Beet root, Local	30.00	86.76	1.98	0.21	1.41	1.65	7.98	3.31

*Note: CF=Crude fiber, Prot= Protein, Carb= Carbohydrate, RS= Reducing Sugar

Table 4.3: Minerals and vitamin content of 100 g of Potato and Tubers.

Index	Food and Description	Ca mg	Iron mg	P mg	K mg	Na mg	Zinc mg	TC µg	Vit C Mg
1	Potato, MS-42-3	9.21	0.62	55.19	460.25	3.87	0.33	216.67	29.94
2	Potato, <i>Janakdev</i>	9.95	0.54	47.18	461.14	3.95	0.29	186.67	25.64
3	Potato, <i>Khumal upahar</i>	9.37	0.50	45.35	453.83	4.01	0.36	187.33	26.30
4	Sweet potato, <i>Suntale 1</i>	31.40	0.26	35.43	345.75	24.46	0.14	7978.67	22.63
5	Sweet potato, Bensishar white	24.89	0.30	35.40	346.26	25.75	0.16	396.00	10.56
6	Sweet potato, <i>Kimichaur seto</i>	25.97	0.41	36.87	347.99	26.52	0.16	424.67	10.23
7	Elephant yam, Elephant yam	42.87	0.64	33.13	494.91	15.61	0.28	527.00	16.49
8	Chayote root	34.71	1.12	62.01	379.62	12.63	0.15	491.33	6.95
9	Taro	53.56	0.87	52.16	435.31	12.29	0.19	292.67	13.77
10	Cassava	20.66	0.49	43.11	353.46	10.15	0.38	312.67	10.24
11	Beet root, Local	17.25	0.72	37.45	313.19	62.19	0.31	20.00	5.38

*Note: Ca= Calcium, P= Phosphorus, K= Potassium, Na= Sodium, TC= Total Carotenoids, Vit C= Vitamin C

Chapter 5

Nutritional Profile of Fruits

5.1 Fruits: Product Information

Fruits are the mature ovary of flowering plants, which normally consist of seeds. Fruits are classified into simple fruits (develop from a single ovary of one flower, grapes), aggregate fruits (develop from multiple ovaries of different flowers, raspberries), multiple fruits (develop from ovaries of multiple flowers growing in a cluster, pineapple), and accessory fruits (include other flower parts in addition to ovary, strawberries). Fruits are a good source of natural sugar, dietary fiber, minerals, vitamins, and antioxidant compounds. Fruits are generally consumed fresh in the kitchen, whereas they are processed to make juice, dried fruits, wine, jam, etc. They are semi-perishable so require specific storage conditions.

Table 5.1: Physical properties of Fruits.

No.	Food and Description	Scientific name	Collection area	Physical	Edible/inedible part	
				L/B Ratio	Edible part (g/100g)	Inedible part (g/100g)
1	Pear, Sandy pear	<i>Pyrus pyrifolia</i>	<i>Kathmandu</i>	0.79	69.52	30.48
2	Sweet orange, <i>Sthaniya junar</i>	<i>Citrus sinensis</i>	<i>Sindhuli</i>	0.95	69.46	30.54
3	Sweet Orange, Washington novel	<i>Citrus sinensis</i>	<i>Dhankutta</i>	1.01	76.62	23.38
4	Mandarin orange, <i>Khoku 1</i>	<i>Citrus reticulata</i>	<i>Dhankutta</i>	0.89	65.63	34.37
5	Mandarin orange, 95 OKEJU	<i>Citrus reticulata</i>	<i>Dhankutta</i>	0.78	71.30	28.70
6	Apple, Fuji	<i>Malus pumila</i>	<i>Jumla</i>	0.84	90.97	9.03
7	Apple, Golden	<i>Malus pumila</i>	<i>Jumla</i>	0.89	89.08	10.92
8	Mango, <i>Dashera</i>	<i>Mangifera indica</i>	<i>Sarlahi</i>	1.55	54.29	45.71
9	Mango, <i>Bombay</i>	<i>Mangifera indica</i>	<i>Sarlahi</i>	1.35	51.19	48.81
10	Lychee, Seedless	<i>Litchi chinensis</i>	<i>Bara</i>	1.07	79.15	20.85
11	Lychee, Patharia red	<i>Litchi chinensis</i>	<i>Bara</i>	1.16	64.19	35.81
12	Banana, G9	<i>Musa paradisicia</i>	<i>Banke</i>	4.13	29.89	70.11
13	Papaya, Local	<i>Carica papaya</i>	<i>Doti</i>	1.43	86.07	13.93
14	Gooseberry, local	<i>Phyllanthus emblica</i>	<i>Doti</i>	0.90	89.49	10.51
15	Guava, Local	<i>Psidium guajava</i>	<i>Doti</i>	1.18	97.89	2.11
16	Strawberry, Tochimeto	<i>Fragaria ananassa</i>	<i>Nuwakot</i>	1.34	97.84	2.16
17	Watermelon, Local	<i>Citrullus lanatus</i>	<i>Sarlahi</i>	1.04	61.42	38.58



Figure 5.1 Pear, Sandy pear



Figure 5.2 Sweet orange, *Sthaniya junar*



Figure 5.3 Sweet orange, Washington novel

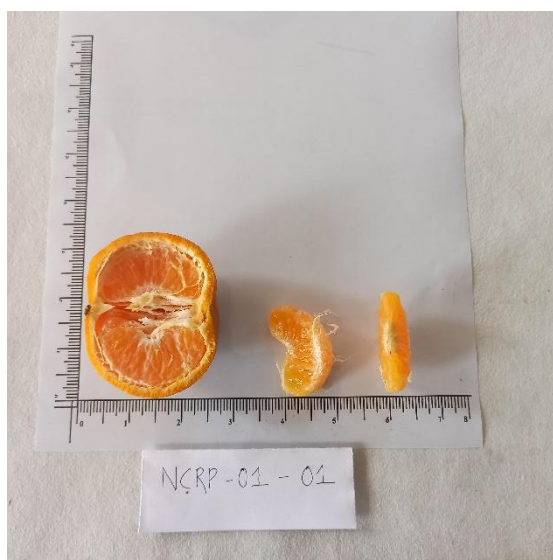


Figure 5.4 Mandarin orange, Khoku 1



Figure 5.5 Mandarin orange, 95 OKEJU



Figure 5.6 Apple, Fuji



5.7 Apple, Golden



Figure 5.8 Mango, *Dashera*



Figure 5.9 Mango, *Bombay*



Figure 5.10 Lychee, Seedless



Figure 5.11 Lychee, Patharia red



Figure 5.12 Banana, G9



Figure 5.13 Papaya, Local



Figure 5.14 Goose berry



Figure 5.15 Guava, Local



Figure 5.16 Strawberry, Tochiotome



Figure 5.17 Watermelon, Local

5.2 Fruits: Nutritional Information

5.2.1: Fruits

Table 5.2: Energy, proximate, and reducing sugar of 100 g of Fruits.

Index	Food and Description	Energy kcal	Water g	Prot g	Fat g	Ash g	CF g	Carb g	RS g
1	Pear, Sandy pear	58.00	84.35	0.51	0.17	0.13	0.19	13.06	4.86
2	Sweet orange, <i>Sthaniya junar</i>	41.00	85.60	0.80	0.09	0.35	1.08	12.07	4.25
3	Sweet Orange, Washington novel	41.00	85.64	0.47	0.10	0.52	1.25	12.02	4.43
4	Mandarine orange, Khoku 1	45.00	87.19	0.57	0.10	0.36	0.61	11.17	3.90
5	Mandarine orange, 95 OKEJU	44.00	86.93	0.99	0.11	0.42	0.74	10.81	3.96
6	Apple, Fuji	55.00	82.22	0.18	0.56	0.37	1.25	15.41	7.35
7	Apple, Golden	49.00	84.43	0.40	0.60	0.25	1.31	13.02	6.53
8	Mango, <i>Dashera</i>	67.00	83.96	0.54	0.20	0.41	0.10	14.79	7.36
9	Mango, <i>Bombay</i>	63.00	84.24	0.57	0.60	0.15	0.17	14.27	6.45
10	Lychee, Seedless	58.00	84.58	0.96	0.24	0.48	0.50	13.24	6.49
11	Lychee, <i>Patharia red</i>	53.00	85.00	0.89	0.25	0.47	0.53	12.86	7.39
12	Banana, G9	101.00	71.01	1.17	0.39	1.07	1.09	25.27	10.65
13	Papaya, Local	21.00	91.56	0.54	0.21	0.58	0.99	6.12	2.75
14	Gooseberry	32.00	81.29	0.46	0.20	0.48	3.76	13.81	2.49
15	Guava, Local	20.00	83.05	1.61	0.38	0.80	4.95	9.21	3.22
16	Strawberry	36.00	87.65	0.95	0.60	0.51	1.28	9.02	2.43
17	Watermelon, Local	18.00	94.71	0.53	0.73	0.16	0.22	3.65	2.63

*Note: CF=Crude fiber, Prot= Protein, Carb= Carbohydrate, RS= Reducing Sugar

Table 5.3: Minerals and vitamin content of 100 g of fruits.

Index	Food and Description	Ca mg	Iron mg	P mg	K mg	Na mg	Zinc mg	TC µg	Vit C mg
1	Pear, Sandy pear	16.79	0.56	3.41	102.79	1.50	0.09	86.67	6.77
2	Sweet orange, <i>Sthaniya junar</i>	40.31	0.59	32.86	160.26	1.30	0.04	854.67	88.91
3	Sweet Orange, Washington novel	38.36	0.46	30.57	161.03	1.56	0.04	890.00	85.76
4	Mandarine orange, Khoku 1	27.19	0.30	17.54	159.72	1.47	0.03	1130.67	59.69
5	Mandarine orange, 95 OKEJU	27.59	0.63	21.32	161.81	1.61	0.05	1028.67	58.99
6	Apple, Fuji	9.95	0.29	9.81	106.23	1.44	0.07	198.78	2.77
7	Apple, Golden	8.31	0.21	8.77	94.71	1.57	0.09	181.00	3.61
8	Mango, <i>Dashera</i>	16.31	0.25	12.73	132.82	1.29	0.97	1483.00	34.15
9	Mango, <i>Bombay</i>	17.44	0.26	10.49	125.39	1.34	0.10	1195.33	27.89
10	Lychee, Seedless	6.76	0.67	25.31	158.36	0.44	0.92	124.00	31.23
11	Lychee, <i>Patharia red</i>	6.28	0.69	23.19	161.15	0.58	0.22	105.33	30.83
12	Banana, G9	7.60	0.19	21.77	333.93	0.29	0.12	235.00	7.77
13	Papaya, Local	14.56	0.20	17.09	170.75	5.34	0.11	2229.67	42.84
14	Gooseberry	19.41	1.19	21.58	231.91	1.42	0.08	61.33	257.19
15	Guava, Local	16.90	0.34	22.59	265.55	2.37	0.21	844.67	213.80
16	Strawberry	16.38	0.36	30.05	145.31	1.16	0.14	243.00	10.37
17	Watermelon, Local	5.45	0.15	10.08	125.83	1.54	0.11	3808.33	46.21

*Note: Ca= Calcium, P= Phosphorus, K= Potassium, Na= Sodium, TC= Total Carotenoids, Vit C= Vitamin C

Chapter 6

Nutritional Profile of Vegetables

6.1 Vegetables: Product Information

Vegetables refer to the various edible parts of a plant and are different from fruits as they are mature ovaries of plants. Vegetables can be classified as leafy vegetables (spinach), root vegetables (radish), stem vegetables (Asparagus), bulb vegetables (onion), flowery vegetables (broccoli), pod vegetables (bean), etc. They are a good source of vitamins, minerals, dietary fibers, and antioxidants. They are considered low-calorie food.

Table 6.1: Physical properties of Vegetables.

S.No.	Food and Description	Scientific name	Collection area	L/B Ratio	Edible part (g/100g)	Inedible part (g/100g)
1	Chilly, <i>Akbare</i>	<i>Capsicum Annuum</i> L'vat cerasiforme	<i>Bagmati</i>	1.03	95.42	4.58
2	Yard Long Bean	<i>Vigna unguiculata</i> subsp.	<i>Bagmati</i>	21.49	99.10	0.90
3	Cucumber	<i>Cucumis sativus</i> L.	<i>Bagmati</i>	4.03	6.54	93.46
4	Pumpkin, Round	<i>Cucurbita moschata</i>	<i>Bagmati</i>	0.88	26.40	73.60
5	Fiddlehead	<i>Matteuccia struthiopteris</i>	<i>Gandaki</i>	67.84	100.00	0.00
6	Pumpkin, Madale	<i>Cucurbita moschata</i>	<i>Bagmati</i>	5.12	99.03	0.97
7	Black mustard	<i>Brassica nigra</i>	<i>Lalitpur</i>	10.75	91.29	8.71
8	Garden cresh	<i>lepidium sativum</i>	<i>Lalitpur</i>	8.35		
9	Spinach	<i>Spinacia oleracea</i>	<i>Lalitpur</i>	4.20		
10	cabbage, green coronet	<i>Brassica oleracea</i> var. <i>capitata</i>	<i>Lalitpur</i>	0.66	85.00	15.00
11	Cauliflower, <i>Khumal jyapu</i>	<i>Brassica olearacea</i>	<i>Lalitpur</i>	0.45	100.00	0.00
12	Cauliflower, Kathmandu local	<i>Brassica olearacea</i>	<i>Lalitpur</i>	0.53	100.00	0.00
13	Raddish, <i>Nino arli</i>	<i>Raphanus sativus</i>	<i>Lalitpur</i>	9.65	82.12	17.88
24	Raddish, <i>Piuthane rato mula</i>	<i>Raphanus sativus</i>	<i>Lalitpur</i>	8.75	82.21	17.79
15	Chayote	<i>Sechium edule</i>	<i>Dhankutta,</i>	1.67	81.64	18.36
16	Bottle gourd	<i>Momordica charantia</i>	<i>Lalitpur,</i>	1.07	94.10	5.90
17	Colocasia tender Leaves	<i>Colocasia esculenta</i>	<i>Lalitpur</i>	79.31	87.56	12.44
18	Mustard green	<i>Brassica juncea</i>	<i>Lalitpur</i>	1.74	100.00	0.00
19	Pumpkin tender leaves	<i>Cucurbita maxima</i>	<i>Lalitpur</i>	53.14	74.69	25.31
20	Chayote tender leaves	<i>Sechium edule</i>	<i>Dhankutta</i>	47.69	95.01	4.99
21	Tomato, <i>Srijana</i>	<i>Solanum lycopersicum</i>	<i>Lalitpur</i>	1.06	99.76	0.24
22	Brinjal, <i>Parwanipur 1</i>	<i>Solanum melongena</i>	<i>Bara</i>	2.99	93.58	6.42
23	Asparagus	<i>Asparagus officinalis</i>	<i>Bagmati</i>	37.56	94.11	5.89
24	Tomato, <i>Khumal 2</i>	<i>Solanum lycopersicum</i>	<i>Bagmati</i>	0.98	99.82	0.18
25	Bitter gourd, <i>patle</i>	<i>Momordica charantia</i>	<i>Bagmati</i>	5.85	98.92	1.08
26	Sponge gourd, New <i>Narayani</i>	<i>Luffa aegyptiaca</i>	<i>Bagmati</i>	12.23	93.18	6.82



Figure 5.18 Chilly, Akbare



Figure 5.19 Yard Long Bean

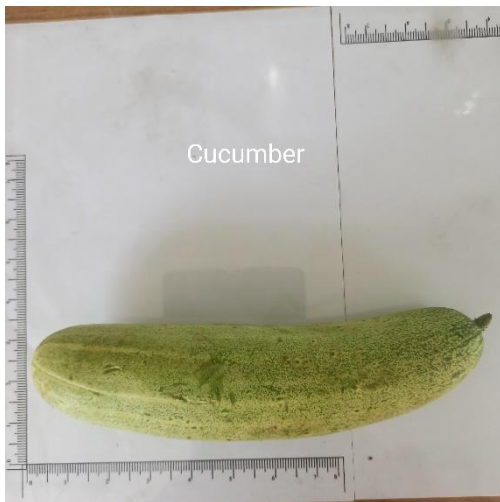


Figure 5.20 Cucumber



Figure 5.21 Pumpkin, Round





Figure 5.22 Fiddlehead



Figure 5.23 Pumpkin, Madale



Figure 5.24 Black mustard



Figure 5.25 Garden cresh



Figure 5.26 Spinach



Figure 5.27 cabbage, green coronet



Figure 5.28 Cauliflower, *khumal jyapu*



Figure 5.29 Cauliflower, *Kathmandu local*



Figure 5.30 Raddish, *Nino aril*



Figure 5.31 Raddish, *Peuthane rato mula*



Figure 5.32 Chayote



Figure 5.33 Bottle gourd



Figure 5.34 Colocasia tender Leaves



Figure 5.35 Mustard green



Figure 5.36 Pumpkin tender leaves



Figure 5.37 Chayote tender leaves



Figure 5.38 Tomato, *srijana*



Figure 5.39 Brinjal, *Parwanipur 1*



Figure 5.40 Asparagus



Figure 5.41 Tomato, *Khumal 2*



Figure 5.42 Bitter gourd, *patle*



Figure 5.43 Sponge gourd, New *narayani*

6.2 Vegetables: Nutritional Information

3.2.1: Vegetables

Table 6.2: Energy, proximate, and reducing sugar of 100 g of Vegetables

Index	Food and Description	Energy kcal	Water g	Prot g	Fat g	Ash g	CF g	Carb g	RS g
1	Chilly, <i>Akbare</i>	52.00	85.45	2.81	0.67	0.62	0.94	9.50	2.34
2	Yard Long Bean	22.00	88.77	2.74	0.16	0.78	1.53	6.01	0.41
3	Cucumber	16.00	95.25	0.37	0.09	0.39	0.45	3.45	1.46
4	Pumpkin, Round	26.00	92.03	1.42	0.11	0.63	0.63	5.17	1.28
5	Fiddlehead	32.00	86.76	4.33	0.34	2.67	1.06	4.84	0.48
6	Pumpkin, <i>Madale</i>	24.00	90.94	1.02	0.05	0.55	0.93	6.52	2.06
7	Black mustard	36.00	90.60	3.39	0.26	2.04	0.74	2.96	0.12
8	Garden cresh	35.00	82.73	4.66	0.38	2.18	2.50	7.55	0.86
9	Spinach	46.00	87.92	2.70	0.26	2.08	0.56	6.48	0.28
10	cabbage, green coronet	22.00	91.22	0.53	0.06	0.53	0.80	6.86	0.86
11	Cauliflower <i>khumal jyapu</i>	45.00	89.99	1.61	0.15	0.75	0.15	7.36	0.49
12	Cauliflower, <i>Kathmandu local</i>	27.00	88.71	1.18	0.47	0.80	1.64	7.20	0.52
13	Raddish, <i>Nino arli</i>	26.00	90.13	0.80	0.05	0.85	1.12	7.05	0.38
14	Raddish, <i>Puthane rato mula</i>	17.00	89.81	1.36	0.06	1.08	1.76	5.93	0.75
15	Chayote	25.00	91.74	1.03	0.06	0.41	0.68	6.08	0.70
16	Bottle gourd	26.00	91.65	0.57	0.07	0.44	0.67	6.59	0.41
17	Colocasia tender Leaves	30.00	84.05	3.08	1.19	2.15	3.01	6.53	0.35
18	Mustard green	16.00	88.12	2.51	0.24	1.73	2.85	4.54	0.09
19	Pumpkin tender leaves	42.00	84.43	4.42	0.67	2.27	1.77	6.44	0.11
20	Chayote tender leaves	20.00	88.88	3.19	0.08	1.56	1.97	4.32	0.14
21	Tomato, <i>srijana</i>	17.00	93.41	1.22	0.17	0.57	1.46	3.16	1.95
22	Brinjal, <i>Parwanipur 1</i>	26.00	89.60	1.56	0.32	0.66	1.20	6.66	0.51
23	Asparagas	19.00	92.42	3.27	0.22	0.59	0.85	2.65	0.48
24	Tomato, <i>Khumal 2</i>	18.00	93.38	0.84	0.25	0.51	0.70	4.32	1.15
25	Bitter gourd, <i>patle</i>	20.00	91.37	1.48	0.29	0.83	1.11	4.92	0.11
26	Sponge gourd, New <i>Narayani</i>	23.00	92.95	1.20	0.17	0.30	0.35	5.03	0.39

*Note: CF=Crude fiber, Prot= Protein, Carb= Carbohydrate, RS= Reducing Sugar

Table 6.3: Minerals and vitamin content of 100 g of Vegetables.

Index	Food and Description	Ca	Iron	P	K	Na	Zinc	TC	Vit C
		mg	mg	mg	mg	mg	mg	µg	mg
1	Chilly, <i>Akbare</i>	14.67	0.49	21.33	337.11	2.31	0.24	2485.33	157.80
2	Yard Long Bean	45.15	1.02	55.52	339.01	10.34	0.59	829.00	31.56
3	Cucumber	11.79	0.57	27.99	185.73	6.17	0.15	160.00	5.93
4	Pumpkin, Round	12.63	0.50	28.71	186.26	5.49	0.12	1116.67	12.61
5	Fiddlehead	36.88	1.41	115.27	418.43	26.35	0.83	3957.33	25.85
6	Pumpkin, <i>Madale</i>	26.31	0.28	27.74	175.25	4.02	0.12	1249.33	13.55
7	Black mustard	56.45	1.09	98.43	627.39	16.04	0.55	4750.67	83.07
8	Garden cresh	204.79	9.57	39.06	362.76	24.16	1.18	4344.33	42.11
9	Spinach	88.96	5.90	38.92	609.38	41.81	0.46	8437.67	47.77
10	cabbage, green coronet	50.63	0.34	38.13	225.09	11.56	0.14	270.00	15.59
11	Cauliflower <i>khumal jyapu</i>	33.94	0.89	40.84	321.00	26.25	0.25	43.00	62.57
12	Cauliflower, <i>Kathmandu local</i>	66.46	0.45	28.79	324.21	29.24	0.39	40.67	70.13
13	Raddish, <i>Nino arli</i>	50.26	0.35	27.45	271.43	20.00	0.18	24.33	12.23
14	Raddish, <i>Puthane rato mula</i>	41.66	0.53	18.67	246.61	19.41	0.17	19.67	15.62
15	Chayote	21.43	0.57	37.29	112.59	1.03	0.09	21.33	9.33
16	Bottle gourd	15.44	0.86	12.61	163.77	1.28	0.16	98.00	12.98
17	Colocasia tender Leaves	218.56	3.85	70.85	398.30	9.90	0.68	25109.00	44.48
18	Mustard green	205.27	9.62	29.14	404.15	18.66	0.62	5489.67	59.89
19	Pumpkin tender leaves	209.89	2.45	72.76	418.80	10.43	0.86	7769.33	11.37
20	Chayote tender leaves	30.17	0.47	25.29	421.41	11.33	0.89	4547.00	26.30
21	Tomato, <i>srijana</i>	12.46	0.27	20.04	155.13	9.92	0.95	4348.67	28.80
22	Brinjal, <i>Parwanipur 1</i>	13.68	0.49	34.85	208.45	3.07	0.29	139.33	2.31
23	Asparagas	31.39	1.83	56.21	211.87	1.85	0.44	3499.00	22.81
24	Tomato, <i>Khumal 2</i>	8.83	0.22	19.18	188.06	9.14	0.10	4682.00	27.12
25	Bitter gourd, <i>patle</i>	18.22	1.17	45.87	317.19	11.47	0.35	452.33	43.86
26	Sponge gourd, <i>New narayani</i>	27.09	0.83	36.73	182.11	7.41	0.29	78.00	7.09

*Note: Ca= Calcium, P= Phosphorus, K= Potassium, Na= Sodium, TC= Total Carotenoids, Vit C= Vitamin C